Jay Niketan Pathare

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Master's student with real-time healthcare and LLM pipeline experience, building AI agents and distributed systems.

EXPERIENCE

Software Engineer Practicum | HeinOnline (University at Buffalo), Buffalo, NY *January 2026 – April 2026*

- Engineered a fault-tolerant metadata extraction pipeline combining CrossRef API, web scrapers, and OCR/LLM vision (Tesseract, Qwen3-VL 8B, Qwen3.5-35B) to process 28k+ pages of law journals.
- Merged all three sources into one validated structure with fuzzy matching, cutting manual data entry by ∼90% and pushing extraction accuracy to 91%+ over five iterations, benchmarked against ground truth.
- Parallelized the pipeline with a CPU-based worker pool that pulls journals from a shared queue, cutting total processing time by over 80%, and added confidence scoring to flag low-confidence fields for human review.

Software Engineer | Thesis Mumbai Tech (Startup), Mumbai, India *August 2024 – August 2025*

- Led end-to-end delivery of healthcare platform modules with a team of 4: gathered requirements in client meetings, prototyped to validate scope, and broke the finalized work into tasks.
- Developed Patient Management and Consent modules on normalized schemas for 10k+ records, plus a real-time WebSocket + Redis pipeline streaming IoT baby-warmer data into PostgreSQL with ∼2s live monitoring.
- Delivered a WebRTC telemedicine feature that let remote doctors review test results with patients and on-site nurses, plus PDF report generation and a color-blindness diagnostic module (95%+ accuracy) with automated reporting.
- Containerized 3+ production projects with Docker Compose for consistent local, staging, and deployment environments.
- Built a heartbeat monitoring dashboard for 4 production projects, pinging each site every 5 minutes to track health.
- Conducted 20+ technical interviews and onboarded new hires, mentoring them through weekly coding rounds over their first 3 weeks and introducing them to the production projects and tech stack.

Cloud Engineer Intern | Data Maven Pvt Ltd, Mumbai, India *November 2023 – May 2024*

- Deployed backend services on AWS EC2 and RDS, using VPC networking and security groups to restrict access.
- Automated cloud resource provisioning, cutting setup time by ∼40% and reducing deployment issues.

Data Engineer Intern | Go Digital Technology Consulting, Mumbai, India *June 2023 – August 2023*

- Developed a Python (Pandas, NumPy) and MySQL ETL workflow to clean and standardize transaction and ledger data and generate automated anomaly reports for the compliance and audit team.

PROJECTS

LedgerFlow: Event-Driven Payment Ledger | Java, Spring Boot, Kafka, PostgreSQL | GitHub *October 2026*

- Implemented a payment ledger in Java and Spring Boot that posts every payment exactly once, even when messages are duplicated or retried, using a transactional outbox and double-entry accounting.
- Stopped overspending under concurrent requests with PostgreSQL row locks: in a 200-request race on an account funded for 50 payments, exactly 50 succeeded and the balance ended at $0.
- Tested with 94 automated tests and a chaos run of 18,000 duplicate events and 14 JVM crashes with 0 violations; a Kubernetes pod-restart demo finished 250/250 payments with no duplicates.

RIPPLE: Agentic Change-Impact Analysis | Python, Pydantic, Git, LLM API | GitHub *August 2026*

- Developed RIPPLE, a Python CLI agent that predicts which files, tests, and regression risks a feature request will affect before any code is written, using 7 static-analysis tools.
- Capped the agent at 25 tool calls and verified every claim in Python; on 30 unseen tasks from 16 repositories (90 runs) it reached 0.306 precision and 0.311 MRR with 0 unsupported claims.

LLM Application Optimization Engine | Python, Qwen (Novita API), Groq API, pytest, LaTeX | GitHub *July 2026*

- Engineered a multi-model pipeline that turns a job description into a tailored one-page resume, with Python checking each generated claim against source evidence and rejecting unsupported ones.
- Cut runtime by ∼73% using three concurrent workers, regenerating only failed parts and fixing issues locally instead of repeating model calls; covered by 2,500+ pytest tests.

EDUCATION

Master's in Computer Science *August 2025 – February 2027*
University at Buffalo, The State University of New York, Buffalo, NY **GPA: 3.85 / 4.0**

Bachelor of Engineering, Information Technology
Vivekanand Education Society's Institute of Technology, Mumbai University, India *August 2019 – May 2023*

TECHNICAL SKILLS

- **Languages & Databases:** Python, Java, JavaScript, Go, SQL, PostgreSQL, MySQL
- **Frameworks & Libraries:** Django, Django REST Framework, Spring Boot, ReactJS, Pandas, NumPy, PyTorch, scikit-learn, Pydantic
- **AI/ML:** LLM Integration, Prompt Engineering, Machine Learning, Deep Learning, Agentic AI, AI Agents, Tool Calling, Model Evaluation, Neural Networks (CNN, GNN), Transformers
- **Software Engineering:** OOP, REST APIs, Full-Stack Development, Backend Development, System Design, ETL, Parallel Processing, Agile, Real-Time Monitoring, Event-Driven Architecture, Distributed Systems, Apache Kafka
- **Tools:** Linux, Redis, AWS (EC2/RDS), Docker, Git, WebSockets, OAuth/Auth0, Kubernetes

CERTIFICATIONS & AWARDS

- **1st Place, ACV Auctions + Copart Challenge** | AI for Good Hackathon Design Challenge, UB *October 2026*
- **AWS Certified Cloud Practitioner** *July 2024*